

KEY FIGURE

MARQUIS DE VAUBAN
1633–1707

France's greatest military engineer, Vauban, directed his first siege in 1657, aged 24. Having won the confidence of King Louis XIV, he oversaw the construction of 37 French fortresses and improved more than 300. But he was, above all, valued for his innovative and aggressive conduct of sieges, employing the system of saps and parallels.



▲ After directing 48 sieges during the course of his career, Vauban retired in 1703 with the rank of marshal.

KEY DEVELOPMENT

THE DEVELOPMENT OF SIEGE WARFARE

By the late 17th century, the construction of fortifications and the conduct of sieges in Europe had become elaborate and formalized, and were the subject of learned treatises by renowned engineers, as well as the focus of military campaigns.

The introduction of cannon into European siege warfare from the 15th century triggered the rapid evolution in defensive fortifications. The challenge was to make a fortress that was less vulnerable to cannon fire than the old stone castles, while being capable of using cannon as a defensive armament. In place of high castle walls, military engineers developed a squat structure known as a “star fort” (see p.117). This was a polygonal fortification, half buried behind a deep, wide ditch, with bastions protruding at each angle. The bastions were wedge-shaped artillery platforms that provided an all-round field of fire, so that soldiers assaulting the walls, or another bastion, would come under fire from the flank or rear. A slope, or “glacis”, of earth in front of the ditch further protected the walls against cannon shot. This basic model was soon developed further, with outworks – fortifications built outside the fortress to delay or prevent besieging forces advancing their guns and soldiers to the main walls.

NEW STRUCTURES

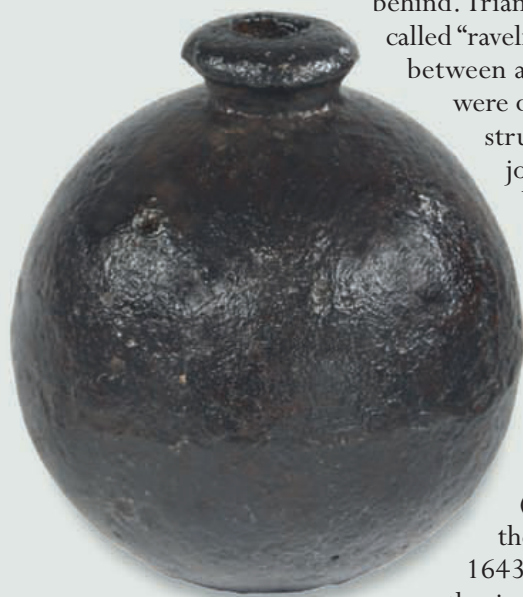
During the 17th century, the complexity of major fortifications left the simple “star fort” model behind. Triangular fortifications, or outworks, called “ravelins” (or “demi-lunes”) thrust forward between angle bastions. Vulnerable points were defended by “hornworks” – defensive structures comprising two bastions joined by a short wall. Citadels were built as back-up forts within fortress towns, where garrisons could continue resistance after the town had fallen. These complex fortifications were expensive to build and man. Many fortifications were built by the Dutch Republic, which was threatened first by Spain in the Eighty Years’ War (1568–1648) and then France in the wars of Louis XIV (reigned 1643–1715). Louis XIV, meanwhile, authorized his chief engineer, the Marquis de Vauban, to build a string of fortresses along the frontiers of France.

By the time of Louis XIV’s reign, sieges were the focal points of conflicts. In the Nine Years’ War (1688–97), pitting France against a Grand Alliance

that included the Dutch, there were 21 major sieges. Field battles resulted chiefly from efforts to relieve ongoing sieges.

WEAPONS OF WAR

Special weaponry evolved for siege warfare, including mortars that launched explosive shells over walls or into siege trenches, and grenades, which were hurled by a new type of elite soldier, the grenadier. Engineers, such as Vauban and his Dutch counterpart, Menno van Coehoorn, formalized and set out the conduct of sieges in treatises. Despite this formality, the siege tactics



▲ MORTAR SHELL

Mortars fired explosive shells consisting of a spherical iron casing packed with gunpowder and ignited by a burning fuse. They saw much use during sieges.

